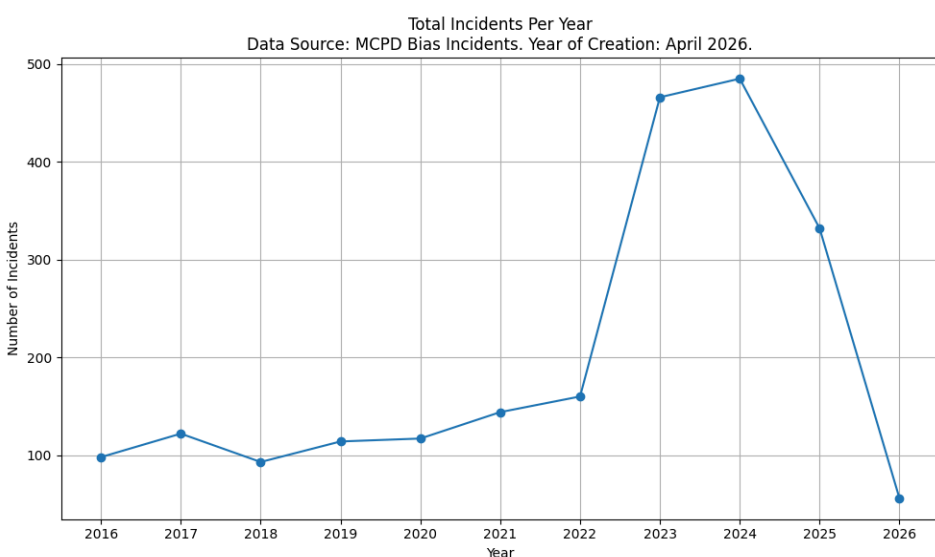


Final Capstone Project

Jacob Dawson
Professor Mais Alraee
Data 205
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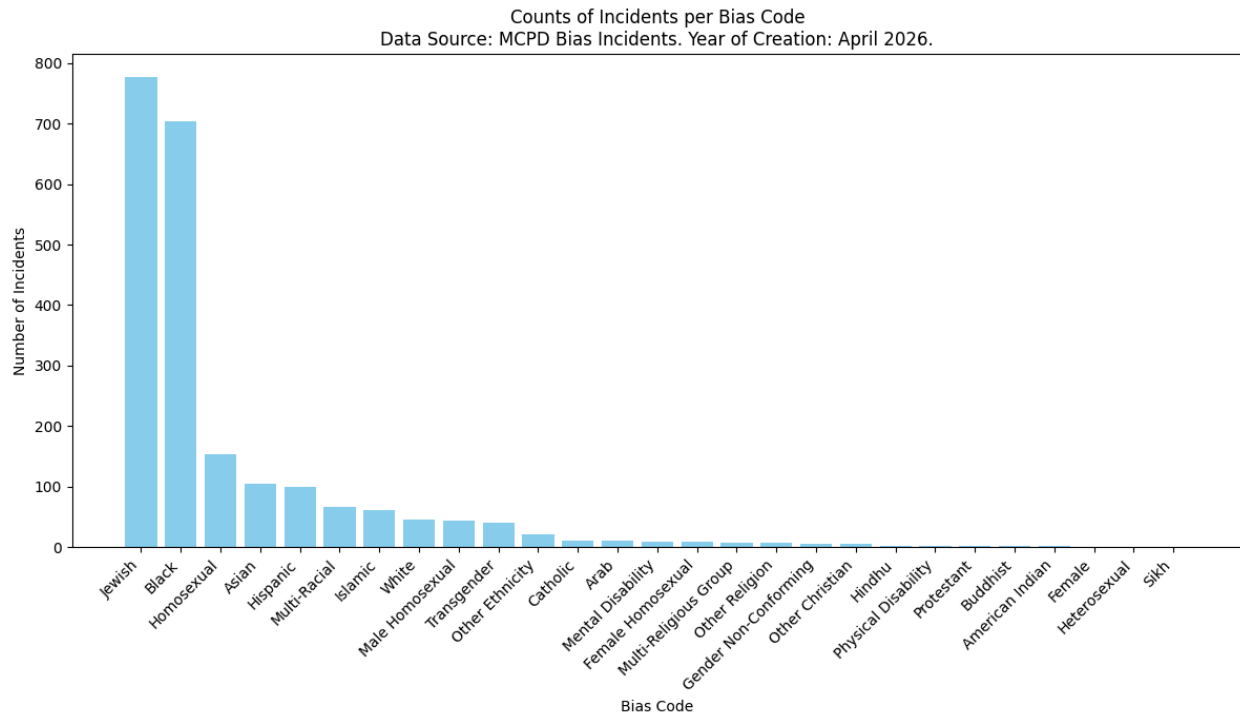
My project utilizes the [MCPD Bias Incidents data set](#) and the up to date Census ACS 5 data starting from 2016 to predict and analyze bias incidents in Montgomery County. According to The Oregon Department of Justice, “A bias incident is any hostile expression that may be motivated by another person’s actual or perceived protected class, meaning their race, color, disability, religion, national origin, sexual orientation, or gender identity.”¹. The goal is to determine what features of cities in Montgomery County cause bias incidents, and then use those features for prediction through modeling. On top of this, through analysis, EDA, and using API links to keep data and notebooks up to date; people can access and view the project in years to come as it remains relevant. The environment I use is [Google Colab](#) which utilizes python as its source code. I also use many libraries such as pandas, seaborn, matplotlib, scikit-learn, statsmodels, and more. Lastly, my method of retrieving data is using API source code for both the MCPD Bias Incidents dataset and ACS 5 data. This method ensures my notebooks remain relevant and up to date with data. Through this, my main analysis and model making is curated via a cleaned version of the MCPD Bias Incidents dataset and the combination of the MCPD Bias Incidents dataset with demographic data from ACS 5 grouped by city and year. The steps for ingestion are repeated throughout my entire project for consistency and time relevancy.



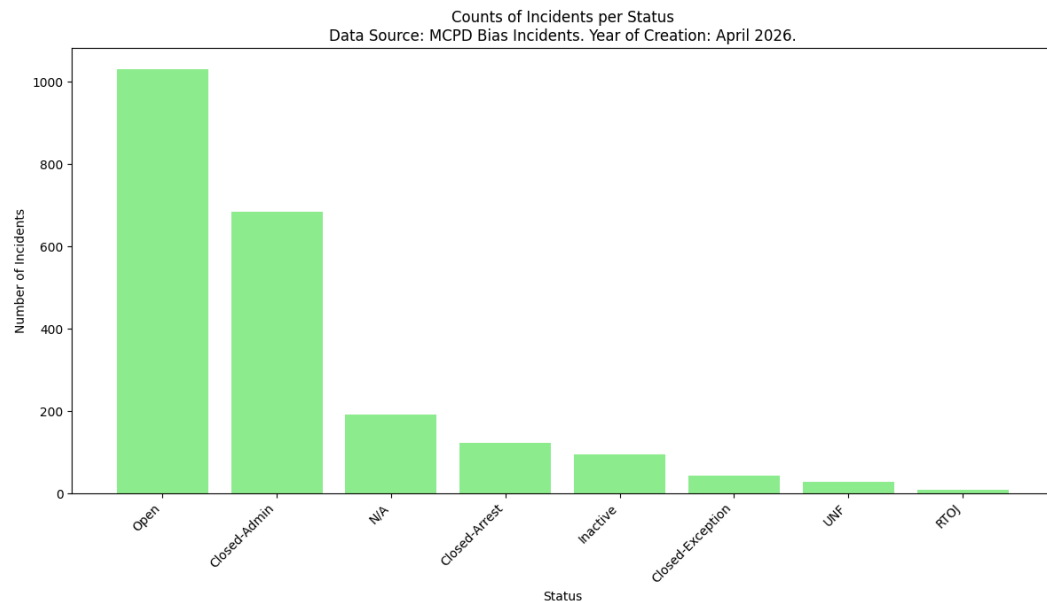
The following is a brief summary of statistics via visualizations. From data of 2016 to 2026, there was a sudden spike of incidents in 2023 that continued in 2024. This is most likely due to the October 7th attack which ignited the Israeli and Palestinian conflict. When I create my model it will reflect this pattern via years as well as demographic

¹ Oregon Department of Justice, “What’s the Difference Between a Hate or Bias Crime and a Bias Incident? - Oregon Department of Justice.”

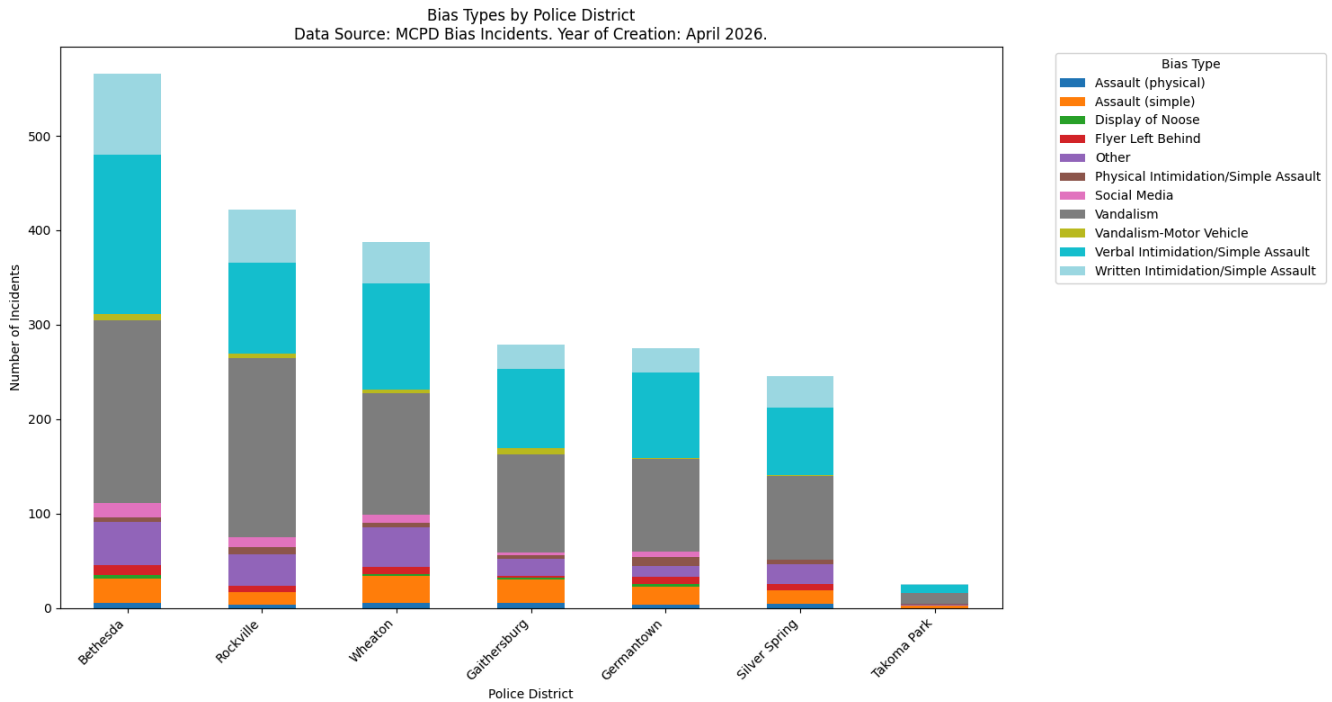
features.



This histogram shows the counts of incidents per race/ethnicity/religion. It is extremely skewed toward anti Jewish and Black incidents which is where I found the most difficulty in my project. This was pretty surprising to me as I thought it would be slightly more evenly distributed with other groups having higher counts. This barchart in green shows the counts of the different statuses for cases. Most of the cases (over 1000!) are open, meaning they have not found any suspects and have not solved the bias incident. This is a huge issue as most cases should not be open like this, preferably at least half of the cases should be closed for a 50% success rate.



Lastly, this stacked bar chart shows the counts of incidents per city in Montgomery county as well as the distribution of crimes per city. Good news is the distribution is normal across all cities, meaning specific crimes are not associated with cities. However, curtain cities do have higher counts which can be predicted as due to



population size. All of these summary statistics are the building blocks to my analysis conducted which is mainly hypothesis testing. Important to note is because the census ACS5 dataset currently only goes up to 2024, my analysis and modeling is from 2016 to 2024.

#1 Z-test for Resolution Rates: Religious vs. Racial/Ethnic Biases:

Null Hypothesis (H0): There is no difference in the resolution rates between religious and racial/ethnic bias incidents.

Alternative Hypothesis (H1): There is a statistically significant difference in the resolution rates between religious and racial/ethnic bias incidents.

Finding: There is a statistically significant difference (Z-statistic: -1.98, P-value: 0.0479). Racial bias incidents (40.81% resolved) are more likely to be resolved than religious bias incidents (36.22% resolved).

#2 Chi-square Test (Fisher's Exact Test) for Religious Bias Groups vs. Resolution:

Null Hypothesis (H0): There is no association between religious bias groups and their resolution status.

Alternative Hypothesis (H1): There is a statistically significant association between religious bias groups and their resolution status.

Finding: There is no statistically significant association (P-value: 0.2235). The likelihood of resolution does not significantly differ between Anti-Jewish incidents and other grouped religious biases.

#3 Chi-square Tests for Racial/Ethnic Bias Groups vs. Resolution (Including Anti-American Indian):

Null Hypothesis (H0): There is no association between racial/ethnic bias groups and their resolution status.

Alternative Hypothesis (H1): There is a statistically significant association between racial/ethnic bias groups and their resolution status.

Finding: A statistically significant association was found (Chi2: 25.69, P-value: 0.0003). However, this finding should be interpreted cautiously because the “Anti-American Indian” category had very low counts, violating chi-square assumptions.

#4 Chi-square Tests for Racial/Ethnic Bias Groups vs. Resolution (Excluding Anti-American Indian):

Null Hypothesis (H0): There is no association between racial/ethnic bias groups (excluding Anti-American Indian) and their resolution status.

Alternative Hypothesis (H1): There is a statistically significant association between racial/ethnic bias groups (excluding Anti-American Indian) and their resolution status.

Finding: There is a statistically significant association (Chi2: 25.58, P-value: 0.0001). The type of racial/ethnic bias significantly impacts whether an incident is resolved.

#5 Z-test for Under-18 Suspects in Bias Incidents:

Null Hypothesis (H0): The proportion of under-18 suspects among all suspects with known age is less than or equal to 50%.

Alternative Hypothesis (H1): The proportion of under-18 suspects among all suspects with known age is greater than 50%.

Finding: There is a statistically significant result (Z-statistic: 10.70, P-value: 0.0000). Individuals under 18 are disproportionately more likely to be suspects in bias incidents (65.26% of known-age suspects).

#6 Pearson’s Correlation: Median Household Income vs. Average Annual Bias Incidents:

Null Hypothesis (H0): There is no linear correlation between Median Household Income and Average Annual Bias Incidents ($\rho = 0$).

Alternative Hypothesis (H1): There is a linear correlation between Median Household Income and Average Annual Bias Incidents ($\rho \neq 0$).

Finding: There is a statistically significant positive correlation (Pearson Corr: 0.86, P-value: 0.0269) between Median Household Income and Average Annual Bias Incidents.

#7 Pearson’s Correlation: Total Population vs. Average Annual Bias Incidents:

Null Hypothesis (H0): There is no linear correlation between Total Population and Average Annual Bias Incidents ($\rho = 0$).

Alternative Hypothesis (H1): There is a linear correlation between Total Population and Average Annual Bias Incidents ($\rho \neq 0$).

Finding: No statistically significant linear association was found (Pearson Corr: -0.42, P-value: 0.4102).

#8 ANOVA: Mean Annual Bias Incidents Between Cities:

Null Hypothesis (H0): The mean annual bias incidents are equal across all cities.

Alternative Hypothesis (H1): At least one city has a different mean annual bias incident rate than

the others.

Finding: There is a statistically significant difference in the mean annual bias incidents between cities (F-statistic: 2.59, P-value: 0.0361). At least one city's mean annual bias incident rate differs significantly from the others.

#9 Pearson's Correlation: Yearly Resolution Rate vs. Yearly Population Proportion:

Null Hypothesis (H0): There is no linear correlation between yearly resolution rates for racial bias incidents and the yearly population proportions of the corresponding racial groups ($\rho = 0$).

Alternative Hypothesis (H1): There is a linear correlation between yearly resolution rates for racial bias incidents and the yearly population proportions of the corresponding racial groups ($\rho \neq 0$).

Finding: No statistically significant correlations were found for any racial bias type tested, including Anti-Black, Anti-White, Anti-Asian, and Anti-Hispanic incidents.

#10 Pearson's Correlation: Average Resolution Rate vs. Average Population Proportion:

Null Hypothesis (H0): There is no linear correlation between average resolution rates for racial bias incidents and the average population proportions of the corresponding racial groups ($\rho = 0$).

Alternative Hypothesis (H1): There is a linear correlation between average resolution rates for racial bias incidents and the average population proportions of the corresponding racial groups ($\rho \neq 0$).

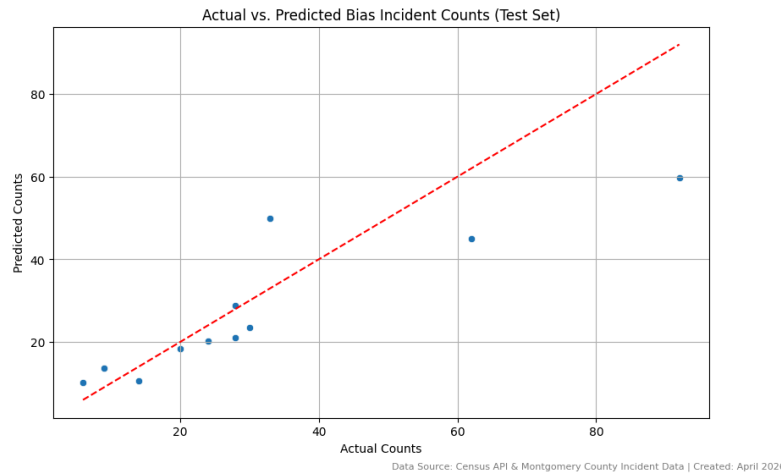
Finding: No statistically significant correlations were found between average resolution rates and average population proportions for Anti-Black, Anti-Asian, Anti-White, or Anti-Hispanic incidents, although Anti-Asian incidents approached statistical significance (Pearson Corr: 0.77, P-value: 0.0723).

The analysis found that racial/ethnic bias incidents were resolved at significantly higher rates than religious bias incidents. While no significant relationship was found between religious bias subgroups and resolution status, significant associations were identified between specific racial/ethnic bias categories and whether incidents were resolved. The results also showed that individuals under 18 were disproportionately represented as suspects in bias incidents. At the city level, median household income was positively correlated with average annual bias incidents, while total population was not significantly related to incident frequency (unlike what you would guess from EDA!). Finally, cities differed significantly in their average annual bias incident rates, but no significant correlations were found between racial population proportions and resolution rates for racial bias incidents.

My modeling focuses on predicting incidents over time and whether a case will be resolved or not. This is done via random forest regression, decision tree, and logistic regression. By using the year variable we are able to see the trend over time and hypothesize what future years may look like. The issue is the sudden spike in 2023 and 2024 was due to a factor outside of the variables in the dataset (Israeli & Palestinian conflict) so the model will more than likely over estimate years after 2024. In fact, it had only about 55% accuracy for the year 2025.

1. Bias Incident Count Prediction (Random Forest Regressor)

Model Performance: The Random Forest Regressor achieved an R-squared value of 0.72 and a Root Mean Squared Error (RMSE) of 12.67 (calculated from an MSE of 160.57). This indicates that the model explains approximately 72% of the variance in the bias incident counts, and the average magnitude of the errors in predicting incident counts is around 12.67 incidents, suggesting a reasonably good fit for predicting the number of incidents based on demographic features and the incident year.

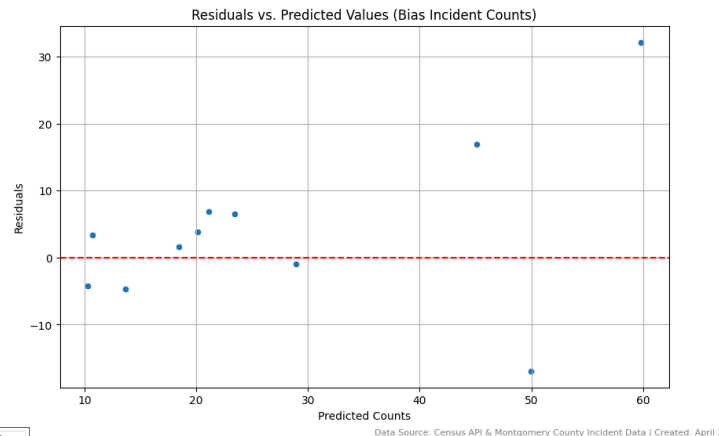


Visual Analysis (Actual vs. Predicted Counts):

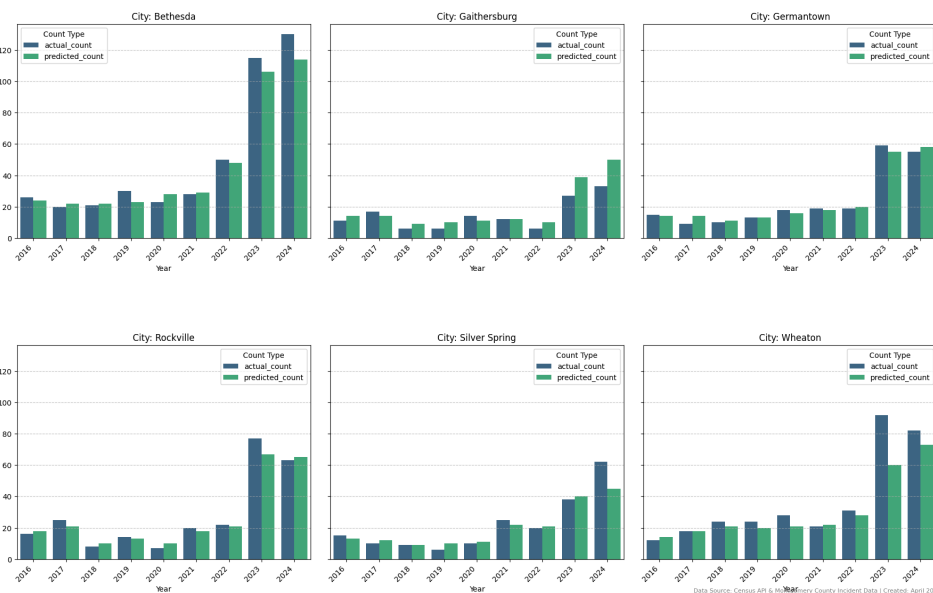
The scatter plot of actual vs. predicted counts showed a positive correlation, generally indicating that as actual counts increase, predicted counts also tend to increase. However, there was notable scatter, especially at higher incident counts, suggesting the model can sometimes underpredict or overpredict significantly for

individual data points.

The residual plot displayed a somewhat scattered pattern around zero, without a strong, clear non-linear trend, which is a positive sign. However, some spread in the residuals was observed, which could indicate slight heteroscedasticity or room for improvement by



Actual vs. Predicted Bias Incident Counts (2016-2024)



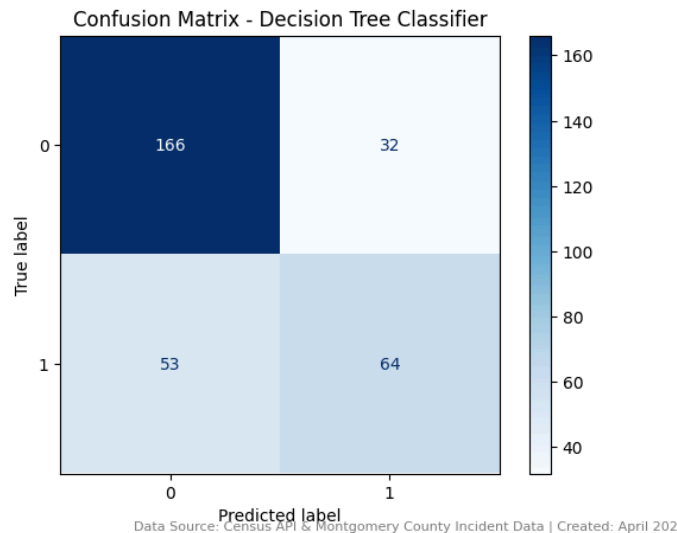
including additional features or more complex model architectures. The city-wise bar plots comparing actual and predicted incident counts over the years (2016-2024) revealed that the model generally captures the overall trends in incident counts for most cities. However, it often smooths out the peaks and valleys, meaning it might struggle to accurately predict extreme fluctuations or specific outlier years

within certain cities. Cities like Bethesda and Rockville showed a relatively good alignment between actual and predicted values, while others might exhibit larger discrepancies in specific years.

2. Case Resolution Prediction (Decision Tree & Logistic Regression)

Decision Tree Classifier: Accuracy: 0.73, ROC AUC Score: 0.72

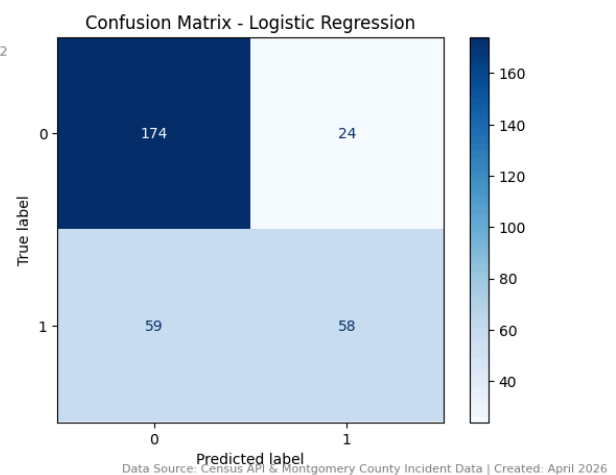
Logistic Regression: Accuracy: 0.74, ROC AUC Score: 0.74



Both models demonstrated strong performance in identifying unresolved cases (class 0), with high recall (0.84 for Decision Tree, 0.88 for Logistic Regression) and good precision (0.76 for Decision Tree, 0.75 for Logistic Regression). This means they are quite effective at correctly predicting when a case will not be resolved.

However, both models struggled more with identifying resolved cases (class 1), exhibiting lower recall (0.55 for Decision Tree, 0.50 for Logistic

Regression). This suggests that a significant portion (around 45-50%) of actually resolved cases were incorrectly predicted as unresolved (false negatives). The precision for resolved cases was moderate (0.67 for Decision Tree, 0.71 for Logistic Regression), meaning that when they did predict a case as resolved, they were correct a decent percentage of the time.



This project shows that bias incidents in Montgomery County are influenced by more than just population size and that demographic and yearly trends can help explain changes in incident counts and resolution rates. The analysis found significant differences between different types of bias incidents, especially when comparing racial/ethnic incidents to religious incidents, while also showing that under 18 suspects make up a large portion of known offenders. My modeling was able to predict general trends fairly well, but sudden world events like the Israeli and Palestinian conflict created spikes that are difficult for models to fully account for. One of the most important parts of this project is that the notebooks and datasets can continue updating over time through APIs, meaning the analysis can stay relevant years from now instead of becoming outdated. This hopefully serves as a guideline for future researchers in Montgomery County who wish to do their own analysis on bias incidents for years to come.

References

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